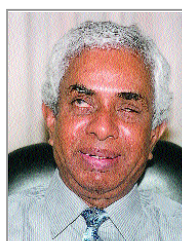


features ■ firsthand

transparent architecture to support either of the disabilities is inherently present in most technologies. Simple magnification methods present in Windows such as increasing the font sizes and using special, high-contrast colour schemes help the partially blind.

Among the enhanced accessibility features in Microsoft's Windows 2000 and XP are two utilities called Narrator and On-Screen Keyboard, which along with the previously introduced Magnifier, are installed by default with the operating systems. They appear along with the Accessibility Wizard and the new Utility Manager in the expanded Accessibility



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GENERAL MANAGER, IDBI

“Braille was inadequate for my mathematical and statistical needs”

menu, found under the Start button. Such utilities provide a minimum level of functionality for many people with disabilities, without requiring additional software and hardware.

While Magnifier, Narrator and On-Screen Keyboard do not replace full-featured accessibility aids, they allow users with disabilities to effectively use a computer running Windows. Narrator is particularly useful to the visually impaired as it is a text-to-speech program that can narrate as a user types. The On-Screen Keyboard displays a virtual keyboard on the screen, enabling users with limited mobility to input text using a pointing device like the mouse. The Magnifier, as suggested by the name, enlarges sections of the computer screen for increased read-

ability, a boon for users with weak or failing eyesight.

However, both the Magnifier and the Narrator are but assisting tools that can not offer the user complete independence. The Magnifier helps only those with failing vision, while the Narrator, by its maker's own confession, “allows only a minimal level of functionality for users with special needs”. The situation demands utility programs with advanced features that let users perform day to day tasks with complete independence.

A glimmer of light

Many a surprised colleague has stopped by Raghavan's desk, kindly letting him know that his monitor is switched off as he types furiously away at his keyboard. Raghavan explains patiently that he doesn't need the monitor to be switched on since he uses a software called Window-Eyes. Developed by GW Micro, this is a screen navigating software that reads out explicitly what is presented on your monitor display—default buttons, spaces, line breaks and punctua-

tion included. A series of configured keys let you choose a particular line to be read out slower or even spelt out. The rest of the functions, such as file saving, changing its location, and maximising and minimising a window can also be configured as per the user's convenience and preference. The pace of reading can be set to a speed comfortable to the user.

Raghavan got through his BCom years with the help of traditional human readers and writers during examination and by studying with the aid of audiotapes recorded by his mother. The vacations preceding his final year took him to the US where he was introduced to various OCR software and to Window-Eyes. By the time Raghavan returned, he was happily rattling away at his keyboard while the speakers prattled

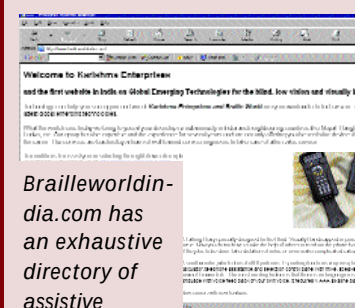
Hardware Solutions

Apart from software enablers, there are hardware devices that let a visually impaired person use the computer in as natural and effortless a way as possible. Devices that give the users the ‘feel’ of the screen through various products such as refreshable Braille displays, touchpads, screen reading software or speech synthesisers, are all

aimed to provide such an experience. But such devices are not easily available and may not match the budget and preference of the user. Adapting them to an already set infrastructure like say, a work place where the basic work devices such as the printers and PCs are connected to each other, becomes quite a task—to install a

Assistive Devices

Karishma Enterprises, run by Ra Agarwal, who is also visually impaired, specialises in importing assistive devices. The company's Web site, www.brailleworldindia.com lists the different devices that can be ordered, and also has a job directory and a space for those



interested in setting up computer and electronic training institutes and projects. Karishma Enterprises undertakes the setting up and installation of these centres as well as the training and orientation of the staff. It also provides consu-

out the words. Since Raghavan was accustomed to using the PC before losing his sight, the screen narration emanating in an American drawl from his speakers/headphones as he used Window-Eyes was easy to decipher.

While the thought of travelling between two continents to learn computing seemed impractical to most visually impaired people, help is available at home. Karishma Enterprises in Mumbai is the master distributor in India for JAWS for Windows (www.freedomscientific.com), a popular and user-friendly screen reading software. It enables the visually impaired to learn and operate a computer independently.

JAWS for Windows, developed by Freedom Scientific USA, has the reputation of being a pioneer in its field. Ramakrishna was one of the early users of JAWS for Windows, used a DOS-based version as early as 1986. The software reads information from the computer screen and speaks back via a speech synthesiser. The popularity of the software is its intuitiveness and ease of use. This is further enhanced by the fact that most of the people working on the project are visually impaired. The s-